Alleviating Space Constraint of Natural Walking in VR and Haptics Retargeting

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Natural walking, one of many navigation techniques, is regarded as the most natural one with the greatest sense of presence. One problem faced by natural walking is the size limitation of physical space. Researchers have developed navigation techniques to enable natural walking in a larger virtual space. In addition, redirection techniques can be utilized to relax the space constraints. Another arduous problem of providing haptics feedbacks for VR experience also exists, some approaches to which share many similarities with redirection techniques. This article seeks to 1. present an overview of current available navigation techniques related to the problem of physical space constraint, redirection techniques and haptics retargeting techniques; 2. highlight their strenghs and limitations; 3. outline the connections between them.

1 Introduction

1.1 Space Constraint

A category of techniques relax the space constraint by scaled translational gain i.e. to scale the velocity of user in physical space [1–4]. The most intuitive and straightforward approach is to scale user's velocity by a constant factor. However, with constant factor, the head sway is also scaled up and can cause cybersickness with high scaling factor [2]. Moreover, Murillo et al also showed that users perform significantly worse in precise control of the viewpoint [4]. There is yet another issue associated with scaled translational gain called drifting-effect [4, 5] which shortens the lifetime of the virtual experience.

Other techniques addressing the problem of space constraint include redirection [6, 7] and the one leverages change blindness [8]. The former manipulates the path of the user in physical space and the latter manipulate the virtual environment for the user to generate desired path. Both techniques have a common goal: to keep user within the boundary of available physical space.

1.2 Haptics Retargeing

Current approaches for providing haptics feebacks in virtual reality can be categorized based on the type of props being utilized. Active props involve robotic actuators or human actors. In contrast, passive props are static physical objects. While active props are often expensive and inconvenient to deploy, providing haptics feedback with passive props is cheaper but challenging and limited. An immediate solution is to place physical props in line with the scenario in virtual environment. However, this solution is limited by the amount and type of physical props available and is inconvenient to deploy if the amount of objects is large. Moreover, this solution cannot be applied to more dynamic virtual environment since the placement of props must reflect every changes to the virtual scene. Another approach, haptics retargeting, redirect users' hands to reach physical objects so that the difference in location between virtual object and physical object can be tolerated. Despite having different purposes, Haptics Retargeting share many simlarities with aforementioned redirection techniques in their implementa-

tions, and thus they are discussed alongside with the redirection techniques.

1.3 Visual Dominance and Change Blindness

In order to redirect user to keep them within the tracked physical space in an imperceptible fashion, two perceptual phenomenon, visual dominance and change blindness, are exploited. With visual dominance, when one perceives conflicted sensory cues, there's a tendency that the the visual cues are trusted, and with change blindness, one tend not to notice the change of scene while his or her vision field is being occluded. Visual dominance are often exploited to scale up or scale down users' turning movements or velocity without them noticing [6,7]. On the other hand, change blindness is leveraged to manipulate the environment surrounding the user for them to generate the desired path [8]. These two phenomenon are also utilized for haptics cues in VR [9–12]. However, both visual dominance and change blindness have their limitations. Furthermore, applications of techniques relying on these phenomenon often impose some extra constraints on the design of the virtual environment.

2 Scaling

2.1 Homogeneous Scaling

Each of the scaling techniques have an associated scaling policy determining what scaling factor is applied at any given time. Homogeneous scaling is the most straightforward policy, the scaling factor of which is a constant and can be described by Eqn.1 [5].

$$\mathbf{P_V}(t_1) = \mathbf{P_V}(t_1 - \Delta t) + \mathbf{k} \odot (\Delta \mathbf{P_R}(t_1)) \quad (1)$$

$\mathbf{P_V}(t_1)$ represents the position in virtual space at time t_1, $\mathbf{P_R}(t_1)$ represents position in real space and $\mathbf{k} = [M, 1, M]^T$ represents the scaling factor, where M is constant and no scaling is applied to the y-axis.

The strengths of homogeneous scaling are that, it is easy to implement and the area of navigable space can be simply computed by scaling area of available real space. However, complications arise if a large scale factor is adopted for homogeneous scaling. First, with large scale factor, the slight motion of head sway will be scaled up causing the viewpoint to move left and right (Fig.1A & Fig.1B). Interrante et al. conducted a user study and collected the user ratings of homogeneous with scale factor={1,10} [2]. The results indicated that participants had worse user experience with scaling factor=10. Second, precise control of the viewpoint becomes much more difficult. Murillo et al. conducted another user study in which participants were required to adjust their viewpoint to indicated position and orientation and the performance of the participants were measured during the process [4]. The same task was repeated with scale factor={1, 2, 4, 8} and the results revealed that using high scaling factor had a profound negative impact on both performance and user experience.

2.2 Seven League Boots

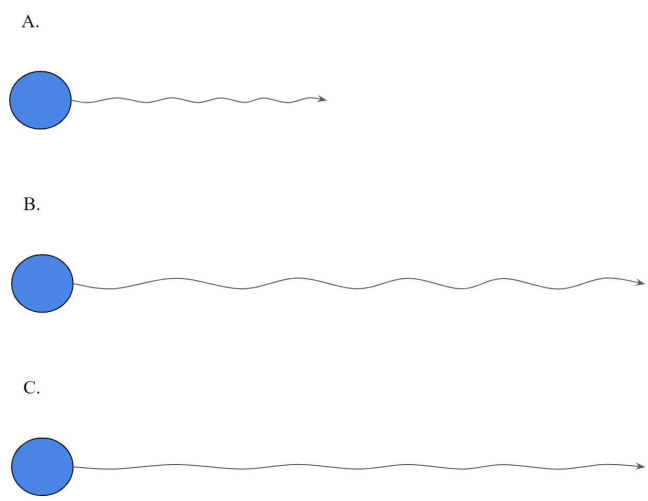

Fig. 1. Illustration of user trajectory. A. Trajectory in real space. B. Trajectory in homogeneous scaling with scaling factor=2. C. Tranjectory in 7 league boots

Interrante et al. proposed the technique called Seven League Boots that effectively mitigates the problem of scaled head sway [2]. This technique determines the direction in which the user is walking and only scales the component of displacement in physical space parallel to that direction (Fig.1C). This technique can be described by

Eqn.2 and Eqn.3.

$$\Delta \mathbf{P_R}(t_1) = \mathbf{V}_{\parallel}(t_1) + \mathbf{V}_{\perp}(t_1) \quad (2)$$

$$\mathbf{P_V}(t_1) = \mathbf{P_V}(t_1 - \Delta t) + \mathbf{k} \odot \mathbf{V}_{\parallel}(t_1) + \mathbf{V}_{\perp}(t_1) \quad (3)$$

The displacement $\Delta \mathbf{P_R}(t_1)$ are decomposed into two components $\mathbf{V}_{\parallel}(t_1)$ and $\mathbf{V}_{\perp}(t_1)$ where $\mathbf{V}_{\parallel}(t_1)$ is parallel to the determined direction and $\mathbf{V}_{\perp}(t_1)$ is perpendicular to that.
In order to determine the direction, the average direction of motion of past few seconds is utilized when the user is walking. If the user was stationary then the gaze-direction is used instead. The experiment results showed that application of Seven League Boots lead to significant improvements on participants' ratings of "easy to use", "feels natural" and "gives accurate impression of distance" while it is less inducing of cybersickness.

2.3 NaviField

Murillo et al. proposed another technique called Navifield to enable precise control of viewpoint [4]. NaviField recognizes the needs for precise maneuvering in certain areas and scaled traveling in others. It defines a set of interactive areas $\mathbf{I}$, in the center of which no scaling is applied to improve performance on maneuvering. Each area $\mathbf{i} \in \mathbf{I}$ is defined as a tuple $\{i_W, r_i, R_i, M\}$, where i_W is the center of two concentric circles of radii r_i and R_i. No scaling is applied within the inner circle and scaling factor of value M is applied outside of the outer circle. The scaling factor between two areas is interpolated based on user's distance from the center i_W. If multiple interactive areas are defined and these areas overlap, the smallest scaling factor will be adopted. A user study is performed to compare NaviField and homogeneous scaling with different average scaling factor S={1,2,4,8}. The results proved that the application of NaviFields can profoundly increase precise control performance and avoid introducing detrimental effects on their subjective experiences.

2.4 Drift-correction

The two techniques described above, Seven League Boots [2] and NaviField [4], used a variable scaling factor instead of constant. They are called scale adaptive navigation and can be described by the Eqn.4 [5].

$$\mathbf{P_V}(t_1) = \mathbf{P_V}(t_1 - \Delta t) + \mathbf{k}(t_1) \odot (\Delta \mathbf{P_R}(t_1)) \quad (4)$$

The constant $\mathbf{k}$ is replaced by a function $\mathbf{k} : R \mapsto R^3$; this function represents the scaling policy e.g. the scaling factor applied in NaviField is dependent on the position in virtual space and varies with time.

An inherent problem of scale adaptive navigation is the drifting-effect [4, 5]. Drifting effect occurs when a user ended up in a different position in real space when he/she finished a closed path in the virtual environment. The amount of displacement is defined as the 'drift'. Drifting effects are undesired because the user will eventually reach the boundaries of the tracked space if a closed path is repeated and reduce the lifetime of the virtual experience. Montano et al. proposed a general framework for SATs (Eq.4) and means to estimate drift by simulating a direct path from the current position to the starting point in VE (since drift is only defined in the starting point in VE) [5]. Based on the framework and the estimated drift, two approaches were developed and tested. Both correction approaches, angular correction and derivative correction, add an additional coefficient, $\mathbf{c}(t) \in [1-\varepsilon, 1+\varepsilon]$, to the scaling factor (Eqn.5).

$$\mathbf{P_V}(t_1) = \mathbf{P_V}(t_1 - \Delta t) + \mathbf{c}(t_1)\mathbf{k}(t_1) \odot (\Delta \mathbf{P_R}(t_1)) \quad (5)$$

The testing showed that both approaches were capable of reducing the magnitude of drift, but, angular correction outperformed derivative correction significantly. Further study also suggested applying angular correction can increase the lifetime of both NaviFields [4] and Seven League Boots [2] without impairing user experience.

2.5 Summary

NaviField [4] and Seven League Boots [2] address two different problems of homogeneous scaling. NaviField enables more precise control of

view point, whereas, Seven League Boots solves the problem of scaled head sway. Both NaviField and Seven League Boots confronted another difficulty called drift-effect [4], causing reduction of lifetime. Drift-correction technique [5] can mitigate the drift-effect. Besides, it can be conveniently applied to general SATs by using the framework proposed alongside with the drift-correction technique. Furthermore, NaviField, Seven League Boots and drift-correction can be applied in combination via the framework to achieve multiple goals at once. Drift-correction [5]

3 Redirection

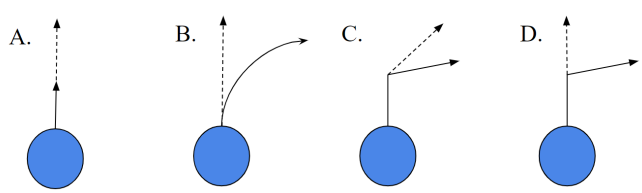

Fig. 2. Illustration of user trajectory. Dotted lines represent trajectory in virtual space. Solid lines represent trajectory in real space

Redirection techniques manipulates the user's path in the real space to keep user within the the boundaries. A typical approach is to inject rotational distortion to the virtual environment i.e. to rotate the virtual space around the user. Distortion can be injected when the user is traveling, so that the user walks a curved path while believing themselves to walking straight in the virtual space (Fig.2B); distortion can be injected when the user turn direction in order to 'steer' the user away from the boundaries in the real space (Fig.2C).

3.1 Redirected Walking

Razzaque et al. developed a technique called Redirected Walking that redirect users relying on their insensitivity to rotational distortion [6]. The technique was tested and found out to be capable of enabling users to travel in a virtual space larger than the real space without them noticing any distortions. The participants had no previous knowledge of the lab size before and were surprised by the actual size of the lab when they removed the head mounted devices. Furthermore, Razzaque et al. suggested a 'resetting' technique to prevent users to reach the boundaries, users are distracted, asked to stop and turn their head left and right in order to provide opportunities for the algorithm to inject rotational distortions. After 'resetting', the user can turn themselves away from the boundary while believing they did not change direction (Fig.2D). Such technique can effectively reorient the user and was used in latter researches [3, 7, 9]. The biggest limitation of Redirected Walking is that the users cannot travel freely in the virtual space. They were required to walk between waypoints, stop at each waypoint and turn to next waypoint before start approaching. This limitation is due to the fact that this technique requires the user to perform the turning action to inject sufficient amount of distortion to keep the user within the boundaries.

3.2 Resetting

Redirected Walking [6] excels in redirecting subtly, yet, it does not allow users to freely roam the virtual space because turning actions are required to inject enough rotational gain and it cause conflicts in vestibular cues and visual cues. The resetting technique used by Razzaque et al. for failure prevention [6] was latter utilized to develop other navigation techniques [3, 7] and haptic retargeting technique [9]. A typical approach is to instruct the users to look at a moving "distractor" and reset when the user is about to reach the boundary. It ensures sufficient amount of rotational gain for redirection, so the users are allowed to roam freely without waypoints. The frequency of distractor appearances is dependent on the available real space.

3.3 Leveraging Change Blindness for Redirection

Suma et al. proposed an intriguing redirection technique that leverages change blindness [8]. Instead of injecting rotational distortion as Redirected Walking does [6], it changes the spatial structure of the virtual environment. As mentioned above, changes happened to the scenario when one's vision is occluded tend not to be noticed. In this regard, Suma et al. designed a room

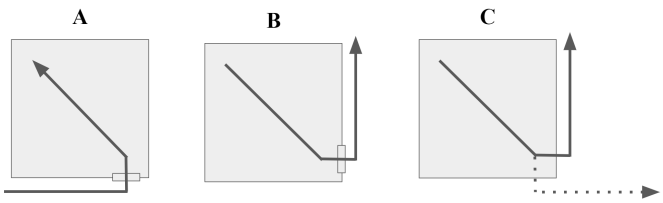

Fig. 3. Illustration of user trajectory. A. Entering the room. B. Leaving the room. C. Dotted line represents the path user believe themselves are taking; Solid line represents their actual path

that changes its structure; the door changes its position and orientation when the user entered the room (Fig.3A & Fig.3B). This design tricked users into thinking they are walking along a long corridor when they are turned 90 degree in real space (Fig.3C). In order to move users' vision away from the door and apply the change, users are required to interact with an object placed at the corner of the room. Suma et al. used this room as a building block and designed a 60'x60' virtual space. The user study indicated that participants can successfully visit each room in the virtual space with only 14'x14' real space. Despite that most participants realize they are moving in circles, only 1 out of 77 was able to recognize the change of position and orientation of the door. This technique is very effective in terms of the amount of virtual space it enabled the users to explore in a more subtle manner without introducing conflicts between vestibular cues and visual cues as Redirected Walking does [6]. However, Suma et al. pointed out that this is not a generalizable technique. First, this technique imposes a great constraint on the structure of the geometry space. Second, the user cannot travel freely in the virual space, they need to enter the room in specific order, otherwise, they will reach the boundary.

3.4 Summary

Redirected Walking [6] imperceivably manipulates the users' walking path by injecting rotational distortion to the virtual environment. However, it requires user to walk a specific path defined by waypoints.

Resetting techniques [3, 7] are essentially variants of redirected walking, they both scale up or down rotations in order to inject rotational distortion. Unlike redirected walking, it allows free roaming by reorienting users before they reach boundary. However, intervention could cause user experience to deteriorate.

Redirection that exploits change blindness changes the structure of building to manipulate his/her path. Unlike the other two, this technique cannot be generalized because it limits the design of virtual space.

Technique	Distortion	Limitations
Redirected Walking	Rotational	Waypoints needed; Causes cues conflict;
Resetting	Rotational	Obvert redirection; Impairs orientation; Intervenes users;
Change Blindness	Architectual	Waypoints needed; structural constraints;

Table 1. Redirection Techniques

4 Haptic Retargeting

Similar to redirection techniques, haptic retargeting techniques exploit visual dominance and change blindness; they redirect user's hand by manipulating position of the virtual representation of the hand [9–11, 13]. Redirection techniques have a common goal–to enable larger virtual space with limited real space, in contrast, researchers utilized Haptic Retargeting to achieve various goals. Haptic Retargeting is utilized to provide feedback for: multiple virtual objects using only one physical prop [9], various scenes using a same general prop [10] and virtual object using physical prop having different shape [11].

4.1 Haptic Retargeting

Azmandian et al. developed and tested three approaches in order to dynamically retarget user's hand to a single passive props to provide haptic feedbacks for multiple virtual objects [9].

The first approach, **Body Warping**, applied incremental translation to the virtual representation of user's hands, the amount of offset between real hand and virtual hand increases as the hand moves

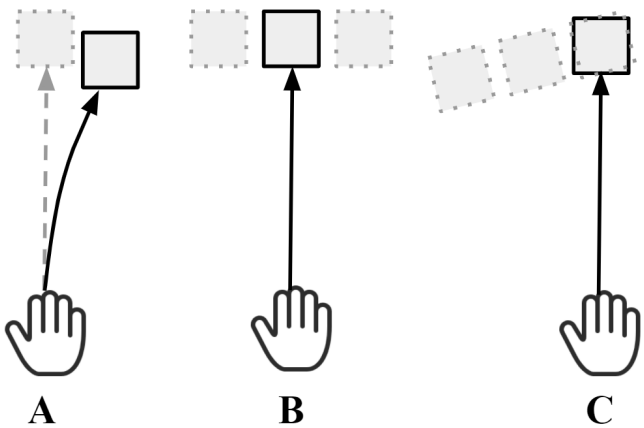

Fig. 4. Illustration of hand trajectory; dotted line represents virtual hand and the solid line represents real hand; dotted box represents virtual object and solid box represents physical object. A. Example of Body Warping. B. Example of World warping. C. Another example of world warping

towards the prop (Fig.4A), so that the offset is imperceivable. This approach requires the target object to be known. Prediction of the target object can be made by comparing the velocity of the hand and the direction vectors of the virtual objects.

The second approach, **World Warping**, injected rotational distortion in a similar fashion with Redirected Walking [6]. The virtual space was rotated to align the target virtual object with the prop (Fig.4B & Fig.4C). Similar to **Body Warping**, the target also need to be known before injecting distortion.

The third approach, **Hybrid Warping**, was a combination of the first two.

Three methods are all tested in a user study, the results of which revealed that, **Hybrid Approach** performed better than the other two approaches in terms of sense of presence, realism and quality of interface. Despite being able to provide haptic feedbacks for multiple virtual objects, this technique is subject to constraints. The users must turn their head left and right for **World Warping** to inject rotational distortion by scaling up or down rotations. Azmandian et al. designed the user study such that the user must turn their heads to read instruction from a billboard on their right hand side before reaching the next virtual object. In addition, the user must move their hand back to the original position before reaching next object.

Fig. 5. General Prop [10]

4.2 Sparse Haptic Proxy

Similar to the body warping approach [9], Sparse Haptic Proxy proposed by Cheng et al. [10] used incremental translation to redirect hand trajectory. Cheng et al. extended haptic retargeting and used a general hemispherical prop (Fig.5). Sparse Haptic Proxy used on-the-fly target remapping allowing user move their hand from one object to another without needing to return their hand to original position. It also introduced eye-tracker for more accurate target prediction. Moreover, it can map target virtual objects to the primitives of the general prop dynamically, allowing the same general prop to be used in different scene.

4.3 Erg-O and 3D retargeted touch

Erg-O proposed by Montano et al. [13] and 3D haptic retargeting proposed by Zhao et al. [11] used optimization to find mappings between positions of the virtual hand and the real hand. Erg-O seeks to find the optimal mapping so that the user can interact with virtual objects in a more ergonomically comfortable position, in contrast,

3D haptic retargeting seeks to warp the proximity space of virtual object to mitigate the mismatch of positions, orientations or shapes between virtual object and its physical prop.

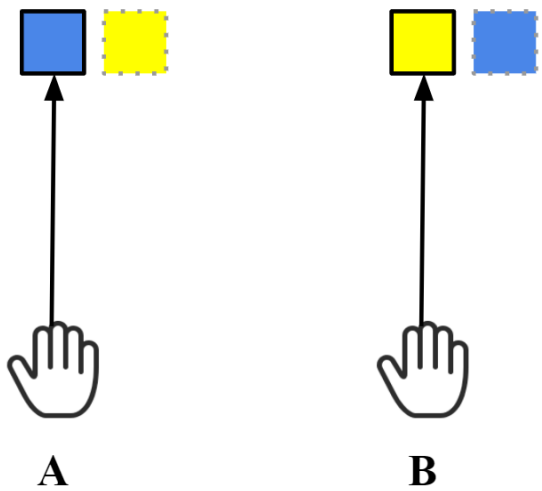

Fig. 6. Change Blindness for haptic

Technique	Approach	Limitations
Haptic Retargeting	Incremental Translation; Rotaional Gain	Return needed; Distractor needed; Prediction needed;
Sparse Proxy Haptic	Incremental Translation	Eye-tracker needed; Prediction needed;
Erg-O	Mapping via optimisation	
3D Retargeted Touch	Mapping via optimisation	
Change Blindness for haptic remapping	Change Blindness	Sequenced target;

Table 2. Haptic Retargeting

4.4 Leveraging Change Blindness for Haptic Remapping

Lohse et al. utilized change blindness for haptic feeback [12]. It swapped positions of virtual objects in order to use only one prop to provide feedback for them. When a user finished interacting with one object aligned with the physical prop (Fig.6A), the object swapped position with the next object so that when the user reached the next object subsequently , he/she actually touched the same prop twice (Fig.6B).

4.5 Summary

Haptic retargeting [9] used hybrid approach combining incremental translation and rotational to redirect users' hands. It was capable to use one single prop to provide feedback for different virtual objects, however, it required the user to return their hand to initial position and a distractor is needed for rotational distortion injection.
Sparse Proxy Haptic [10] extended the incremental translation approach by using general prop (Fig.5), on-the-fly target remapping and eye-tracker for target prediction. These extensions enable the hand to move from one target to another without returning and reusing physical prop.
Erg-O [13] and 3D retargeted touch [11] used optimization to find mapping between position of real hand position and virtual hand position. The benefit is that they do not need target prediction and returning.
Change Blindness for haptic remapping [12] changed the position of virtual objects to align them with the physical props. It required the users to touch virtual object in a specified sequence.

5 Conclusion

This article presented an overview of current approaches for alleviating space constraint of natural walking including scaling and redirection techniques. While scaling techniques can be easily noticed (with high scaling factor), redirection techniques are capable to extend navigable space subtly. Some concepts used for redirection, such as distractor, visual dominance, rotational gain and change blindness, are also utilized for haptic retargeting, for that reason, redirection and haptic retargeting have similar implementations and constraints. In order to identify their common features, this article introduced typical implementa-

tions and outlined their constraints respectively.

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